AI BUILDER CUP 2026

Savvy Cart AI

A grocery plan that accounts for your cards and every extra trip.

Retail & Commerce

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The weekly shopping decision

Karuppan's household

A budget of LKR 12,000, three credit cards and four supermarket chains.

Decision	What changes the answer
What to buy	Purchase history and the next estimated purchase date
Where and when	Shelf prices, store promotions and offer weekdays
Which card	Exclusions, minimum spend, caps and combining rules
One store or several	The saving after the cost of another trip

Karuppan is a fictional Colombo demo persona. Budget and wallet are sample preferences.

Evidence from a household's receipts

26

de-identified Keells bills

21 purchase dates
5 May to 17 October 2026

Receipts show item-level promotions. Four dates contain multiple receipts.

The demo adds 17 fictional bills. Competitor prices and store promotions remain simulated.



Source: samples/purchase-history.json and the redacted receipt. One household, not market validation.

A shopping plan the household can review

1 → **2** → **3** → **4** → **5**

Pantry

Review items and quantities.

Wallet

Choose cards. Set the cost of another trip.

Plan

Compare days, stores and baselines.

Why

Ask questions. Review the tool comparisons.

Approve

Review the plan and mock fulfilment.

Offer Reality Check

Paste bank terms or upload a screenshot to see what the offer pays on this basket.

Implemented guest workflow. The final approval creates a MOCK preview. No order is sent.

Offer Reality Check

Published offer: 10% off on Total Bill every Saturday

Headline arithmetic

LKR 593

Modeled card payout

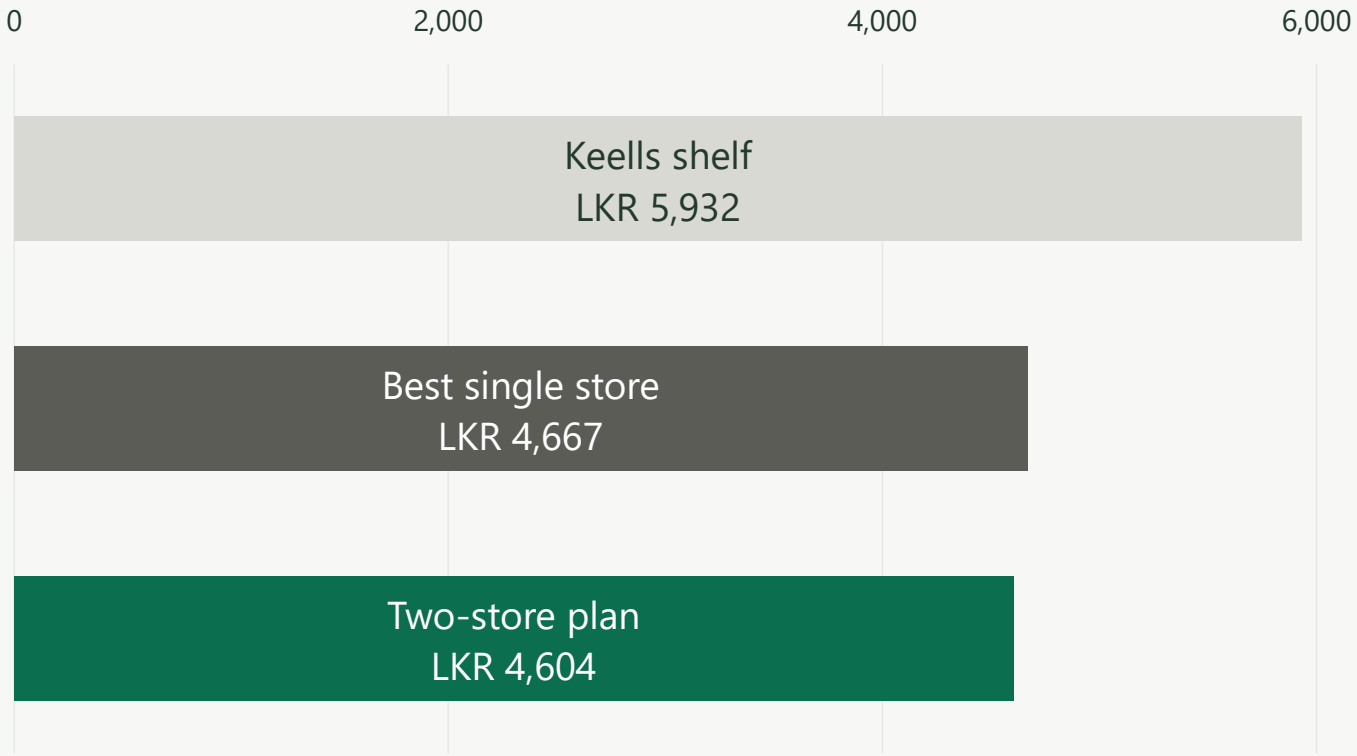
LKR 0

The basket misses the minimum by LKR 2,475

LKR 5,525 bill after store promotions, against an LKR 8,000 minimum.
Six promoted lines cannot combine with this offer.

Seylan Keells terms checked 18 Oct 2026. Sample basket and promotions. No bank affiliation.

The extra trip must earn its cost



LKR 63

extra saving over the best single store

At LKR 320 per extra trip, Cargills alone wins at LKR 4,667.

LKR 1,328 below the shelf baseline. Every saving here is simulated.

Simulated default basket, before adding the sample receipt. Trip estimate LKR 250. Whole-rupee amounts.

Gemini reads the inputs. Code prices the basket.

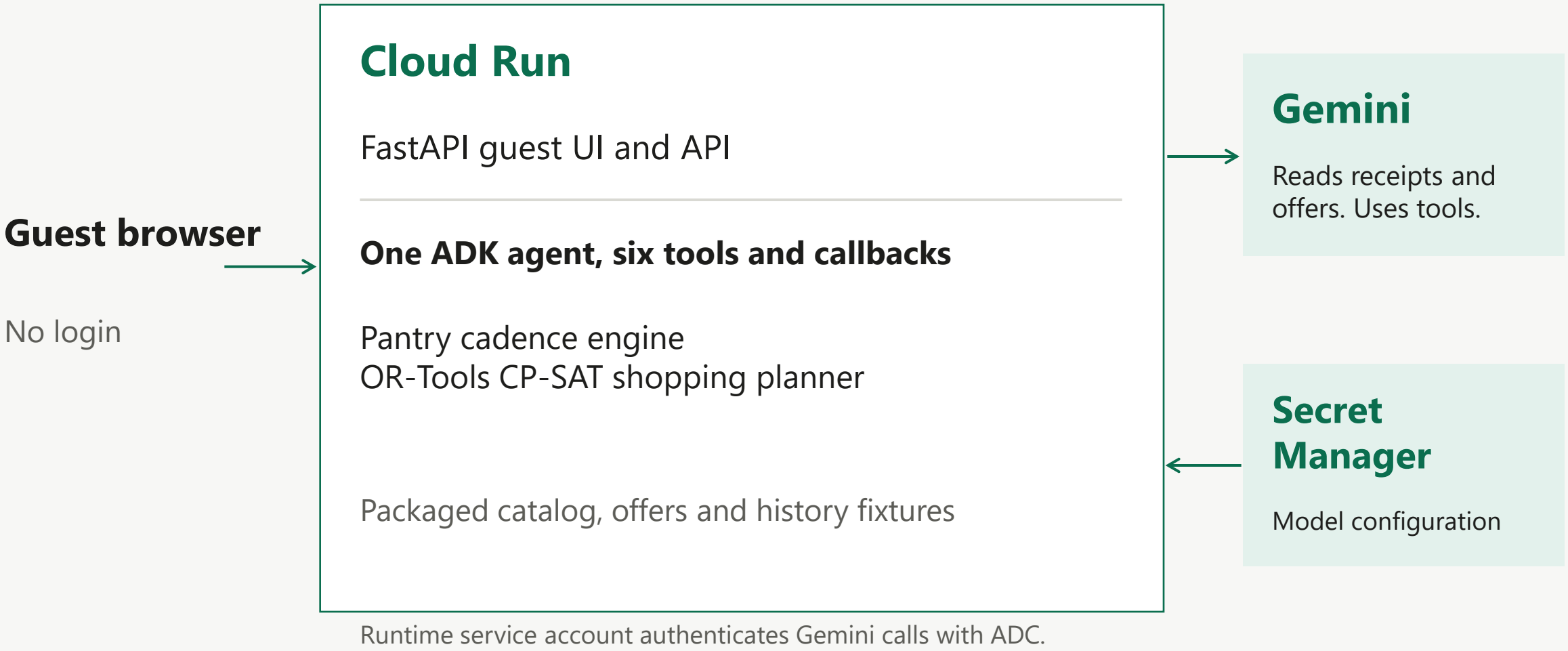
Gemini understands	The application checks and acts
Receipt photo	Catalog SKU, decimal quantity and confidence; update session history
Offer text or screenshot	Structured rate, dates, exclusions, minimum, cap and combining rules
Shopper question	ADK tools compare plans or check an offer against the basket

Shoppers can review and correct extracted terms

The pantry estimates purchase timing. OR-Tools CP-SAT calculates store assignment and cost.

Implemented integration paths. Live Gemini checked 18 Oct 2026: the Why answer about 30 seconds, offer reading about 7 seconds; accuracy sampling is ongoing.

Google Cloud deployment architecture



Deployed on Cloud Run (asia-southeast1, scale to zero). ADK runs inside the application container; no separate Agent Engine deployment.

Implementation evidence and safeguards

120

automated tests pass

Reproducible engine behavior and integration contracts.

Rules in code

Minimums, caps, exclusions, promotion limits and travel cost.

Observable reasoning

Tool arguments and comparisons remain inspectable.

Two guardrail layers

Agent callbacks and page checks filter known numeric failures.

Human decision

No card-number collection. Approval creates a mock preview.

Local test run on 18 Oct 2026: 120 passed. Scripted model tests do not establish live AI performance.

The business model starts with a household pilot

Household value

A plan that shows the conditions and assumptions behind a saving.

Compare the full basket after card rules and travel cost, with reasons for each line.

Distribution hypothesis

Issuers or retailers may support distribution or pay for consented referrals, subject to data agreements.

Ranking follows shopper cost; commercial relationships need disclosure.

Pilot measures

Repeat use and comparison time.
Data freshness and purchased cost versus planned cost.

Proposed validation and commercial hypotheses. No signed partners, revenue or adoption claims.

Colombo first, reusable rules for other markets

1

Household pilot

Validate receipt reading and decisions with real purchases.

2

Authorized data feeds

Add versioned retailer prices and issuer terms with refresh checks.

3

Another market

Adapt catalogs, currencies, rounding and local card rules.

Seeking pilot households and retailer or issuer data partners

Proposed roadmap. Expansion depends on data access, product matching and local offer coverage.

Appendix: prototype boundaries and status

Available now	Still to validate or complete
Live Cloud Run deployment; guest workflow and reproducible totals	Demo video and submission-portal upload
Gemini receipt / offer clients and an ADK tool loop	Live extraction accuracy across many receipts
Real receipt history plus labeled sample data	Fresh retailer prices and complete default offer terms
MOCK approval and example partner APIs	Actual checkout, payments and partner integrations

One household supplies the real history. Savings remain simulated. Guardrails need ongoing evaluation.

Status as of 18 Oct 2026. The deployed Cloud Run link and the public repository are live; the demo video is the remaining submission item.

Appendix: sources and reproducible evidence

Competition requirements and judging

https://aibuildercup.com/themes.html

https://aibuildercup.com/Faqs.html

Published Seylan Keells offer

https://www.seylan.lk/promotions/cards/supermarket/keells

Captured sample: fixtures/v2/offer-sample-seylan.json

Source repository

https://github.com/nyogendran/savvy-cart-ai

Public repository, open to anonymous visitors

Local reproduction

python -m pytest -q

README.md; fixtures/v2; samples/purchase-history.json

External sources checked 18 Oct 2026. Detailed provenance and scenario inputs also appear in speaker notes.